

Deep Sea hydrothermal vent habitats support gravid Antarctic krill

Kim S. Bernard^{1*}, Angelo F. Bernardino², Kristen N. Buck¹, Salvatore Caprara¹, R. Ben Freiberger¹, Elizabeth Hogan³, Rachel L. Kaplan¹, Aaron Micallef⁴, Laura E. Moore¹, Sarah Seabrook⁵, Andrew R. Thurber⁶

Supplementary Materials

Methods

Dive sites and operations

Euphausia superba (commonly name: Antarctic krill; also referred to here as “krill”) were observed and collected via remotely operated vehicle (ROV) during expedition FKt241214 aboard R/V *Falkor (too)* (15 Dec 2024 – 03 Jan 2025). ROV dives relevant to the current study were conducted at Hook Ridge (Dive S0770), a sediment-hosted volcanic ridge in the central Bransfield Strait¹ where two active vent mounds rise 8–12 m above the seabed at 1,030–1,120 m depth, and a contrasting, non-vent-associated site in Antarctic Sound (Dive S0767) (Figure 1). The ROV lights were turned on for the descent of both dives and descent rates averaged 0.34 m s⁻¹ and 0.29 m s⁻¹ for S0770 and S0767, respectively. ROV *SuBastian*, depth-rated to 4,500 m, was equipped with a UHD pan-zoom-tilt science camera (SULIS Z70, 4K resolution) and four auxiliary HD SeaCams. A Sea-Bird FastCAT SBE-49 CTD and an SBE-3 thermistor (recording at 1 Hz, ± 0.002 °C accuracy) were mounted 0.5 m above the skid. A high-temperature PT-100 probe capable of measuring from 0–600 °C was mounted on the starboard manipulator. Navigation was provided by a Sonardyne Sprint-INS, Syrinx DVL, and USBL positioning system. Sampling tools included a five-chamber suction sampler (4 L min⁻¹), two 2-L Niskin bottles, push-cores, and an insulated “biobox”. Illumination was provided by twin 150 W LEDs; total lamp irradiance at 1.0 m was $< 10^{-7}$ W m⁻². Vehicle dimensions 2.7 × 2.2 × 1.8 m; weight in air 3 200 kg; top speed 3 kn (schmidttocean.org).

In addition, during a subsequent R/V *Falkor (too)* expedition (FKt250220; 19 Feb 2025 – 27 Mar 2025) to the South Sandwich Islands, an ROV dive was conducted at a hydrothermal vent field at Quest Caldera (Dive S0804²), located at the northern end of the back-arc spreading East Scotia Ridge³ (Figure 1). The ROV lights were turned on during the descent and the average descent rate was 0.32 m s⁻¹. Video footage from the ~21-hour dive was downloaded and examined by KSB to assess the presence of gravid female Antarctic krill.

Krill collections

Krill were collected during FKt241214 with the ROV *SuBastian* suction sampler on each dive and held in 1-L sample chambers until the ROV was back on deck. All krill remained in good condition and were actively swimming in the sample chambers when the ROV returned.

Demographics of krill from Hook Ridge and Antarctic Sound

Collected krill were first measured for total length (measured from the tip of the rostrum to the end of the uropods; Standard Length¹⁴, then staged for sexual maturity⁵. Following staging, stomachs were carefully, but quickly, dissected out and placed immediately in DNA-RNA Shield for microbial community composition analysis (see details below) and frozen at -80 °C prior to

extraction. Krill bodies were subsequently dried at 60 °C for 24 hours for measurement of dry mass. Only four individuals were collected at Hook Ridge, and all were used for each analysis in the present study. Thirty-five individuals were collected from Antarctic Sound. Of those, four were used for trace metal analysis, five were used to assess gut microbial community composition, and all were used for stable isotope analysis.

Gut and vent-associated microbial community composition: Hook Ridge vs. Antarctic Sound

In addition to the krill stomach samples described above, microbial community compositions were also assessed for rock and bacterial mat samples. Portions of rocks and bacterial mat collected from the hydrothermal edifice, within 25 m of the krill collection (Figure 1) were placed in DNA-RNA Shield. Rocks were differentiated according to their color, including black, white, red, and yellow. All samples (rocks, mat, and krill stomachs) had their DNA extracted using the ZymoBiomics DNA miniprep kit followed by PCR amplification using the Earth Microbiome Project as in Thurber et al.⁶ with updated 515f and 806r primers. Amplicons were sequenced on an Illumina MiSeq (2x250) at the Center for Quantitative Life Sciences at Oregon State University. 16S bioinformatics were carried out using Qiime2 following the data pipeline in Thurber et al.⁶ and through comparison with the Silva V138-99 classifier.

Stable-isotope analysis: Hook Ridge vs. Antarctic Sound

Stable isotope analyses were conducted on bulk whole krill tissue from subsamples (0.7–1.2 mg) of dried and ground krill tissue. Carbon and nitrogen isotopes ($\delta^{13}\text{C}$ vs Vienna-Pee Dee Belemnite, VPDB, and $\delta^{15}\text{N}$ vs air, respectively) were analyzed at the Oregon State Stable Isotope Collaboratory (OSSIC) by continuous flow isotope ratio mass spectrometry using a Thermo-Scientific Flash EA IsoLink inlet connected to a Delta-Q mass spectrometer. OSSIC uses working standards for daily runs that are annually calibrated to international standard reference materials. Carbon isotope data were calibrated with the international standards NIST 8541 (Graphite), NIST 8542 (Sucrose), NIST 8573 (L-Glutamic Acid), NIST 8574 (L-Glutamic Acid) and IAEA 600 (Caffeine). Nitrogen isotope data were calibrated using NIST 8550 (Ammonium Sulfate), NIST 8551 (Ammonium Sulfate), NIST 8573 (L-Glutamic Acid), NIST 8574 (L-Glutamic Acid), IAEA 600 (Caffeine) and IAEA N2 (Ammonium Sulfate). Three working standards are used each run to calibrate $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$. An additional independent working standard with known isotopic composition and a matrix chosen to match the sample matrix as close as possible is run for verification of calibration. No chemical lipid extraction was performed prior to analysis. Instead, $\delta^{13}\text{C}$ values were lipid-normalized post-analysis using the mass-balance correction of Logan et al.⁷.

$$\delta^{13}\text{C}_{\text{corrected}} = \delta^{13}\text{C}_{\text{untreated}} \times 0.967 + 0.861$$

Trace-metal analysis: Hook Ridge vs. Antarctic Sound

Samples were processed for trace metal analyses in a clean room using trace metal clean materials and techniques. Krill digestion procedure was modified after Ratnarajah et al.⁸. Briefly, dried and ground krill tissues were weighed directly into acid cleaned 15-mL Teflon vials (PFA, Savillex), followed by addition of 3 mL concentrated nitric acid (HNO_3 , Optima grade, Fisher) and 0.375 mL concentrated hydrogen peroxide (H_2O_2 , Optima grade, Fisher). Vials were capped, inverted gently several times, uncapped, then placed on a hot plate at room temperature for 20 minutes. Vials were then heated at 125 °C for 8 hours with caps loosely fastened. This was followed by a 5-hour evaporation step at 60 °C to remove any remaining liquid. After vials

cooled to room temperature (2 hours, 20 °C), samples were resuspended in 10 mL of 10% HNO₃ (Optima, Fisher) and inverted gently several times. The same procedure was followed for two certified reference materials – DORM-5 fish protein (n =5) and BCR-414 plankton (n = 6) – and digestion blanks (n =10). Prior to analysis, resuspended samples were diluted in acid-cleaned falcon tubes to achieve 10:1 and 100:1 ratios with 2% HNO₃ (Optima, Fisher). Indium (In) internal standard was added to all sample dilutions to achieve a final concentration of 10 ppb.

The concentrations of phosphorous (P), vanadium (V), chromium (Cr), manganese (Mn), iron (Fe), cobalt (Co), nickel (Ni), copper (Cu), zinc (Zn), arsenic (As), molybdenum (Mo), cadmium (Cd), barium (Ba), and lead (Pb) were analyzed using a high-resolution inductively coupled plasma mass spectrometer (HR-ICP-MS, Element2 ThermoFisher) fitted with a PC³ Peltier Cooling System with Temperature Controlled SSI hybrid dual cyclonic/Scott spray chamber. Samples were analyzed in medium resolution using kinetic energy discrimination (KED) mode to minimize interferences. A nine-point calibration curve (10-10,000 ppt) was prepared from serial dilutions of 1000 ppm certified stock solutions in a 2% HNO₃, 10 ppb In matrix. The calibration curve was extended to 1,000,000 ppt for P. The limit of detection (LOD, Table S2) for each element was calculated as 3 times the standard deviation of 9 reagent blanks (2% HNO₃, 10 ppb In). Samples were corrected via reagent blank subtraction followed by digestion blank subtraction matched by day of digestion and dilution ratio. Only values above the LOD and within the standard curve range for each element are included in the final presented results, with a few exceptions that were above the standard curve range in both the 10:1 and 100:1 dilutions (Supplementary Materials Table S2). Good agreement was achieved between measured and certified values for reference materials DORM-5 (fish protein) and BCR-414 (plankton) for most elements (Supplementary Materials Tables S4 for BCR-414). Greater than 70% recovery was achieved for all elements with a certified value (or indicative reference value) for BCR-414 and 14 of 15 of elements for DORM-5, the exception being Ni at a 45% recovery (Table S4). Fe had a 75% recovery in DORM-5 and 88% recovery in BCR-414, mirroring recoveries reported in Ratnarajah et al.⁸ (DORM-3 79%, BCR-414 87%), who performed a similar digestion protocol. Although Pb and Zn are contamination prone elements, the same sample collection procedures were applied to all samples, and the concentration differences are large enough for us to be confident that the trends are a result of natural processes rather than contamination. All trace metal concentrations were normalized to P as an internal proxy of biogenic material and to account for differences in digestion efficiency between samples.

Quest Caldera Grid Survey

During the dive at Quest Caldera on FKt250220, the ship-board team conducted a 2-hour grid survey around the vent site, covering an area of approximately 300 × 300 m. Video footage from the grid survey was downloaded and manually examined by KSB. Gravid female krill are easily distinguished from males and non-gravid females by their enlarged and rounded thorax. The ripe eggs beneath their carapace give their thorax an opaque appearance, which reflects lights from the ROV more intensely than that of non-gravid individuals. Gravid female krill visible within the ROV field of view were counted at 30-second intervals. At each interval, the number of unique gravid females was recorded over a 4-second video window.

To estimate the density of gravid females along the survey track, counts were converted to individuals per meter traveled. The ROV travelled at an average speed of 0.36 m s⁻¹, so each 4-second counting window represented approximately 1.5 m of forward movement. Counts from each 4-second window were therefore divided by the distance travelled during that window to

standardize them to gravid females m^{-1} . To estimate the number of gravid females between successive 30-second intervals, standardized counts from adjacent time points were averaged and multiplied by the distance travelled between those time points, following a trapezoidal integration approach. Intervals in which the ROV travelled less than 3 m between successive 30-second time-steps were excluded from the analysis as these low-movement intervals were considered susceptible to overestimation of krill abundance. The valid interval estimates were then summed across the full grid survey and divided by the total distance travelled to obtain an overall estimate of gravid female krill per meter of ROV track.

The average ROV speed during the survey was higher than the upper-range spring swimming speed estimated for Antarctic krill of comparable size, approximately 0.26 m s^{-1} for a 48 mm individual based on swimming speeds determined by Kane et al.⁹. This suggests that krill counted at successive 30-second intervals were unlikely to be the same individuals following the ROV lights. This interpretation was also supported by the video footage, in which new individuals could be seen appearing ahead of the ROV as it moved through the survey grid.

Statistics and Reproducibility

Because the sample sizes for trace metal analysis were small ($n = 4$ for each site), we used an exact one-tailed permutation test¹⁰, enumerating all 70 possible reallocations of the eight specimens and defined the p-value as the fraction of the 70 permutations that produced a median difference equal to or greater than the one observed. Ratios were also $\log_{10}$ -transformed so that the difference in site medians could be expressed as a fold-change (10^{Δ}) for ease of interpretation. Next, principal components analysis (PCA) was performed to reduce dimensionality and identify trends in the trace metal dataset, focusing specifically on a subset of eight elements (Pb, Zn, Ba, Cd, Mn, Cu, As, and Cr) relevant to our ecological analysis. Concentrations were log-transformed prior to analysis to normalize data distributions and reduce the influence of extreme values. PCA was implemented using the *prcomp* function from the R stats package (Version 2025.05.1+513), using the correlation matrix (`scale. = TRUE`) to standardize variables with different scales. Differences in metal composition in krill tissue from each site were evaluated with a permutational multivariate analysis of variance (PERMANOVA) using *adonis2*¹¹ in the *vegan* package¹². For microbial community structure analysis, amplicon-sequence variants (ASVs) were log-transformed [$\log_e(x + 1)$] prior to calculating Bray-Curtis dissimilarities and visualizing with non-metric multidimensional scaling (nMDS, $k = 2$). Differences in microbial community structure between sites were assessed using PERMANOVA as described above. An analysis of variance (ANOVA) test was used to compare $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of krill collected at Hook Ridge and Antarctic Sound.

Krill ID	Length (mm)	Sex/Stage	Site
K001	34.06	M3A	Antarctic Sound
K002	29.8	M2A	Antarctic Sound
K003	37.72	M3A	Antarctic Sound
K004	29.62	M2A	Antarctic Sound
K005	36.55	M3B	Antarctic Sound
K006	28.88	J	Antarctic Sound
K007	28.68	J	Antarctic Sound
K008	30.39	F3A	Antarctic Sound
K009	36.21	F3B	Antarctic Sound
K010	31	M2A	Antarctic Sound
K011	31.31	F3A	Antarctic Sound
K012	28.69	F2A	Antarctic Sound
K013	30.82	F3B	Antarctic Sound
K014	37.09	F3D	Antarctic Sound
K015	34.55	M3B	Antarctic Sound
K016	36.7	M3B	Antarctic Sound
K017	29.61	J	Antarctic Sound
K018	26.85	J	Antarctic Sound
K019	33.05	F3C	Antarctic Sound
K020	31.75	M2A	Antarctic Sound
K021	30.75	J	Antarctic Sound
K022	29.76	M2A	Antarctic Sound
K023	32.11	M2A	Antarctic Sound
K024	32.9	F3A	Antarctic Sound
K025	32.66	M2A	Antarctic Sound
K026	26.57	--	Antarctic Sound
K027	31.66	M2A	Antarctic Sound
K028	33.92	M2A	Antarctic Sound
K029	30.77	M2A	Antarctic Sound
K030	32.28	J	Antarctic Sound
K031	27.74	J	Antarctic Sound
K032	29.63	J	Antarctic Sound
K033	30.57	M2A	Antarctic Sound
K034	30.68	J	Antarctic Sound
K035	25.94	J	Antarctic Sound
K036	50.47	F3D	Hook Ridge
K037	48.16	F3D	Hook Ridge
K038	50.37	F3D	Hook Ridge

K039	42.16	F3D	Hook Ridge
------	-------	-----	------------

Table S1. Krill body size, sex, and developmental stage by sampling location. Body length (mm) and sex/stage of krill collected at Antarctic Sound and Hook Ridge. Krill identifiers (Krill ID) match to those seen in the manuscript figures and the supplementary tables that follow. M = male; F = female; J = juvenile. Stages are based on those of Makarov and Denys (1980), reference 2.

Element	LOD (ppt)	Reference LOD mass 50 mg (mg/kg)	Reference LOD mass 100 mg (mg/kg)
As	3.7	0.07	0.04
Ba	3.1	0.06	0.03
Cd	1.2	0.02	0.01
Co	1.0	0.02	0.01
Cr	3.7	0.07	0.04
Cu	6.4	0.13	0.06
Fe	120	2.4	1.2
Mn	2.2	0.04	0.02
Mo	3.0	0.06	0.03
Ni	240	4.8	2.4
P	150	2.9	1.5
Pb	0.6	0.01	0.01
V	2.1	0.04	0.02
Zn	97	1.9	0.97

Table S2. Element-specific limits of detection and reference limits by sample mass. Limits of detection (LOD, mg/kg) for each element, and reference LOD (mg/kg) for two masses (50 and 100 mg).

Element	Concentration (mg/kg)							
	Hook Ridge Krill				Antarctic Sound Krill			
	K036	K037	K038	K039	K003	K014	K015	K016
As	1.86	2.52	3.39	2.51	4.34	6.84	6.81	6.35
Ba	7.02	4.96	8.57	34.61	2.67	3.91	2.25	4.39
Cd	0.20	0.18	0.40	1.00	0.08	0.17	0.07	0.05
Co	0.06	0.09	0.12	0.06	0.06	0.12	0.09	0.18
Cr	0.26	0.38	0.66	0.46	0.67	2.11	4.08	8.75
Cu	42.8	44.6	92.1	130	46.8	75.8	30.4	102
Fe	44.7	45.7	94.3	198	23.0	150	67.3	185
Mn	3.72	3.54	6.27	10.6	1.91	5.92	2.21	5.02
Mo	0.10	0.17	0.17	0.46	0.07	0.12	0.07	0.12
Ni	0.43	0.77	0.81	<LOD	0.78	1.09	1.46	3.48
P	8110	9400	16300	8300	6640	16300	5860	15700
Pb	30.5	4.56	67.0	206	0.05	0.11	0.05	0.11
V	0.17	0.20	0.19	1.04	0.11	0.37	0.31	0.42
Zn	90.4	51.1	184	313	24.4	81.8	27.7	59.6

Table S3. Element concentrations in individual krill from Hook Ridge and Antarctic Sound. Concentrations (mg/kg) of elements in each krill (e.g., K036) from Hook Ridge and Antarctic Sound. Red values note those above the upper range of standard curve.

Element	Dilution	Concentration (mg/kg)	Replicates	Reference (mg/kg)	Recovery (%)
As	10x	6.56 ± 0.51	6	6.82 ± 0.28	96.2
Ba	10x	24.4 ± 2.2	6	ND	ND
Cd	10x	0.29 ± 0.02	6	0.383 ± 0.014	76.9
Co	10x	1.14 ± 0.09	6	*1.43 ± 0.06	79.8
Cr	100x	24.2 ± 1.1	3	23.8 ± 1.2	101.8
Cu	100x	32.4 ± 0.9	3	29.5 ± 1.3	109.7
Fe	10x	1640 ± 130	6	*1850 ± 190	88.4
Mn	10x	267 ± 20	6	299 ± 13	89.4
Mo	10x	1.15 ± 0.15	6	*1.35 ± 0.20	85.1
Ni	100x	13.4 ± 0.7	3	18.8 ± 0.8	71.3
P	10x	15700 ± 1100	6	ND	ND
Pb	10x	3.16 ± 0.28	6	3.97 ± 0.19	79.6
V	10x	7.05 ± 0.72	6	8.1 ± 0.18	87.1
Zn	10x	88.2 ± 6.3	6	111.6 ± 2.5	79.1

Table S4. Recovery of elements from Plankton BCR-414 reference material and quality-control assessment. Recovery of reference material Plankton BCR-414 at 10x or 100x dilution. Mean (± standard deviation) concentrations (mg/kg) of each element are presented for our krill tissue samples. Reference material (shown as mean ± standard deviation, mg/kg) is certified, unless marked with an asterisk (*), which denotes indicative reference material. Percent recovery is calculated as the average concentration of an element in our samples divided by the certified reference value of the same element in the Plankton BCR-414 material, multiplied by 100. Percent recoveries below 70% are marked in red.

Krill ID	Site	%C	$\delta^{13}\text{C}$	%N	$\delta^{15}\text{N}$
K001	Antarctic Sound	44.77	-23.43	10.14	3.62
K002	Antarctic Sound	46.16	-22.37	9.27	2.98
K003	Antarctic Sound	41.91	-23.55	11.12	4.03
K003	Antarctic Sound	41.27	-23.64	11.17	4.11
K003	Antarctic Sound	41.91	-23.80	11.26	3.88
K004	Antarctic Sound	43.26	-23.44	9.44	3.06
K005	Antarctic Sound	43.82	-24.31	10.59	4.43
K006	Antarctic Sound	45.88	-22.71	9.49	3.07
K007	Antarctic Sound	45.14	-23.94	10.52	3.36
K008	Antarctic Sound	42.94	-23.69	8.67	3.50
K009	Antarctic Sound	46.07	-23.08	10.19	2.98
K010	Antarctic Sound	39.57	-22.72	9.95	3.48
K011	Antarctic Sound	43.05	-23.30	10.63	3.51
K012	Antarctic Sound	46.06	-24.02	11.07	2.61
K013	Antarctic Sound	45.32	-23.56	10.73	3.13
K014	Antarctic Sound	43.82	-23.47	10.88	3.01
K015	Antarctic Sound	39.20	-25.55	10.62	3.80
K016	Antarctic Sound	40.56	-23.74	10.23	3.32
K019	Antarctic Sound	44.72	-23.56	10.53	3.60
K020	Antarctic Sound	42.97	-24.23	9.93	3.42
K021	Antarctic Sound	43.32	-22.95	9.62	3.41
K022	Antarctic Sound	42.05	-23.12	8.90	3.25
K023	Antarctic Sound	44.79	-23.10	9.96	2.91
K024	Antarctic Sound	46.94	-23.05	9.32	3.26
K025	Antarctic Sound	48.37	-23.33	9.79	2.75
K026	Antarctic Sound	42.32	-23.71	10.06	3.31
K036	Hook Ridge	45.25	-25.52	11.24	3.15
K036	Hook Ridge	45.78	-25.98	9.63	2.58
K036	Hook Ridge	45.38	-25.63	10.82	2.59
K037	Hook Ridge	39.04	-26.08	10.72	3.36
K038	Hook Ridge	46.21	-26.74	10.14	2.74
K038	Hook Ridge	45.32	-26.49	10.70	2.70
K039	Hook Ridge	41.18	-25.37	10.43	3.16

Table S5. Elemental composition and stable isotope values of krill from Antarctic Sound and Hook Ridge. Elemental and stable isotope analysis results of krill collected at Antarctic Sound and Hook Ridge. Krill identifiers (Krill ID) are provided. Note that duplicates were measured for some krill. Percent carbon and nitrogen (%C and %N, respectively) and $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ are provided.

Supplemental Materials – References

1. Klinkhammer, G. P. *et al.* Discovery of new hydrothermal vent sites in Bransfield Strait, Antarctica. *Earth and Planetary Science Letters* **193**, 395–407 (2001).
2. SOI Data Manager. Dive Reports generated during R/V Falkor (too) expedition FKt250220 (2025). Marine Geoscience Data System (MGDS) <https://doi.org/10.60521/332370> (2025).
3. Leat, P. T. *et al.* Bathymetry and geological setting of the South Sandwich Islands volcanic arc. *Antarctic Science* **28**, 293–303 (2016).
4. Mauchline, J. *Measurement of Body Length of Euphausia Superba Dana*. 9 pp. (1980).
5. Makarov, R. R. & Denys, C. J. *Stages of Sexual Maturity of Euphausia Superba Dana*. 13 (1980).
6. Thurber, A. R., Seabrook, S. & Welsh, R. M. Riddles in the cold: Antarctic endemism and microbial succession impact methane cycling in the Southern Ocean. *Proc. R. Soc. B.* **287**, 20201134 (2020).
7. Logan, J. M. *et al.* Lipid corrections in carbon and nitrogen stable isotope analyses: comparison of chemical extraction and modelling methods. *Journal of Animal Ecology* **77**, 838–846 (2008).
8. Ratnarajah, L. *et al.* Understanding the variability in the iron concentration of Antarctic krill: Understanding iron variation in Krill. *Limnol. Oceanogr.* **61**, 1651–1660 (2016).
9. Kane, M. K., Yopak, R., Roman, C. & Menden-Deuer, S. Krill motion in the Southern Ocean: quantifying in situ krill movement behaviors and distributions during the late austral autumn and spring. *Limnology and Oceanography* **63**, 2839–2857 (2018).
10. Ernst, M. D. Permutation Methods: A Basis for Exact Inference. *Statistical Science* **19**, 676–685 (2004).
11. Anderson, M. J. A new method for non-parametric multivariate analysis of variance. *Austral Ecology* **26**, 32–46 (2001).
12. Oksanen, J. *et al.* vegan: Community Ecology Package. 2.7–1 <https://doi.org/10.32614/CRAN.package.vegan> (2001).